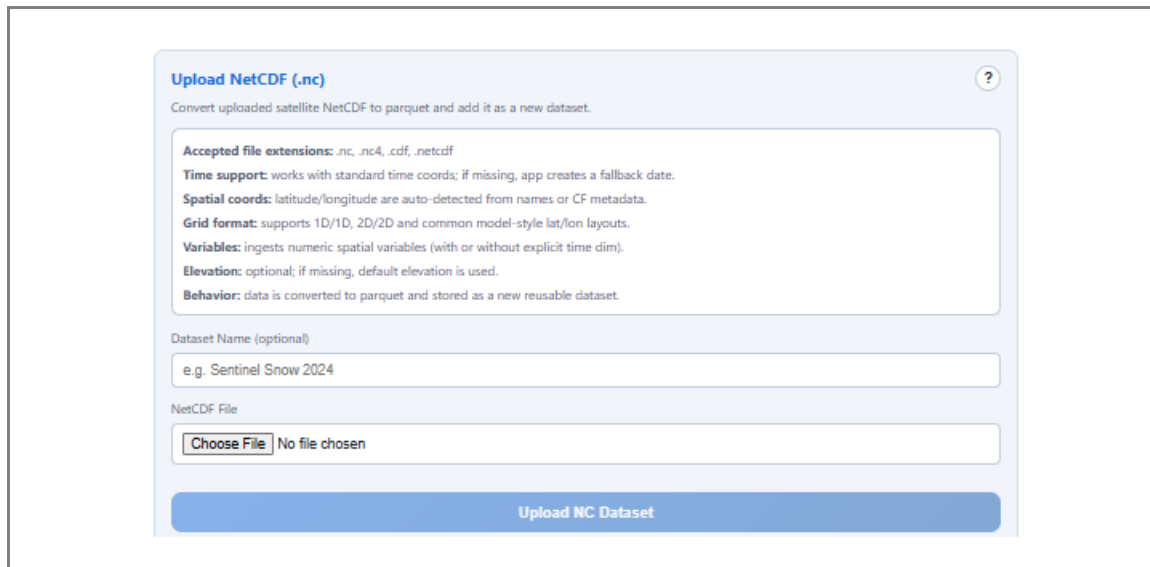


WEEKLY REPORT-5

Completed Tasks:

1. Expanded the platform from a single-dataset workflow to a multi-dataset architecture by adding CMIP6 as a selectable source with its own date range, variables, and file paths.
2. Reviewed and incorporated variable-specific handling for CMIP6, including temperature conversion from Kelvin to Celsius, precipitation conversion to millimeters per day, and radiation conversion to operational analysis units.
3. Added the SPHY model ingestion pipeline by converting GeoTIFF-based layers into parquet files so they can be queried through the same backend structure used for point-style datasets.
4. Integrated glacier shapefiles into the region-selection flow and exposed glacier names where metadata was available so cryosphere-focused subregion analysis can be performed more conveniently.
5. Implemented NetCDF upload support and direct MOD10A1 monthly GeoTIFF support, thereby extending the platform to user-supplied and raster-backed inputs without disturbing the stable core workflows.
6. Continued performance optimization by limiting only the visual density of heavy raster responses while preserving full valid-pixel participation in backend mean calculations.

Figure :



The screenshot displays a web interface for uploading NetCDF files. At the top, it says 'Upload NetCDF (.nc)' with a help icon. Below this, a subtitle reads 'Convert uploaded satellite NetCDF to parquet and add it as a new dataset.' A box contains detailed instructions: 'Accepted file extensions: .nc, .nc4, .cdf, .netcdf', 'Time support: works with standard time coords; if missing, app creates a fallback date.', 'Spatial coords: latitude/longitude are auto-detected from names or CF metadata.', 'Grid format: supports 1D/1D, 2D/2D and common model-style lat/lon layouts.', 'Variables: ingests numeric spatial variables (with or without explicit time dim).', 'Elevation: optional; if missing, default elevation is used.', and 'Behavior: data is converted to parquet and stored as a new reusable dataset.' Below the instructions, there is a text input field for 'Dataset Name (optional)' with the example 'e.g. Sentinel Snow 2024'. Underneath is a 'NetCDF File' section with a 'Choose File' button and the text 'No file chosen'. At the bottom, a large blue button is labeled 'Upload NC Dataset'.

Figure 5. NetCDF integration.

Reporting No: 5 Week No: 5

Date From: 22/06/2026 Date To: 27/06/2026

Student ID: 23DCS076

Internship Title: WebApp for Visualization and Analysis of Himalayan Hydro-Cryospheric Datasets

Plans for next week:

1. Document the computational logic for each dataset and analysis mode inside the application and submission material.
2. Prepare an outcomes section for precomputed scientific products so expensive analyses do not need to be repeated at runtime.
3. Start deployment and offline packaging work so the current system can be shared outside the local development machine.

References:

1. SPHY_Model_data_prep/convert_geotiff_to_parquet.py and related backend integration code.
2. Rasterio, GeoTIFF metadata handling, and NetCDF ingestion references used for format integration.
3. MOD10A1 snow and albedo dataset notes reviewed during dashboard expansion.

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